

Frontier of AI Agent Memory & Thermodynamic Memory Architectures (Mid-2026): A Research Briefing for Ferricula / LUME

TL;DR

- As of mid-2026, agent memory has become a first-class architectural discipline with its own benchmarks (LoCoMo, LongMemEval, BEAM), production frameworks (Mem0, Letta/MemGPT, Zep/Graphiti, A-MEM, Cognee), and a clear convergence on hybrid (vector + graph + lexical) stores with explicit consolidation, decay, and self-organization — exactly the design space the Deep Blue Dynamics "Ferricula" memory engine and "Lume" index occupy.
- The thermodynamic/energy-based framing the project leans on is genuinely a live research frontier: modern Hopfield networks/Dense Associative Memories (DenseAMs) are now formally tied to transformer attention and to diffusion models as energy landscapes, and a January 2026 paper ("Stochastic Thermodynamics of Associative Memory") quantifies tradeoffs between entropy production, retrieval accuracy, and operation speed — directly validating Ferricula's "memory as entropy" metaphor, though the hardware (Extropic TSU, Normal Computing CN101) remains early-stage.
- For the LUME document-index layer, the state of the art is hybrid retrieval (BM25 + dense + graph) with agentic/semantic chunking and late-interaction multi-vector models (ColBERT/ColPali family); Lume's existing BM25 + GTR-T5 + significance-weighted entity-graph blend is well-aligned with the frontier, and the main upgrade opportunities are newer embedding models (Qwen3-Embedding), late-interaction retrieval, and hierarchical/recursive indexing (RAPTOR/HippoRAG-style).

Key Findings

The Ferricula / Deep Blue Dynamics context (located)

The public-facing project is spelled **Ferricula** (the user's "Furricula" with two R's appears to be a variant), hosted under the **DeepBlueDynamics** GitHub organization and at **ferricula.com**. It is described as a "Thermodynamic memory engine for AI agents. Decay, consolidation, dreaming, entropy. Built in Rust" (v0.5.0). Its core conceit: "Ferricula models memory fidelity as an energy state. High-energy memories are Active, but as time passes and disuse grows, they decay naturally unless reinforced. Below the fidelity gate, text is discarded but the vector seed survives — and the synthesizer can reconstruct it." It uses Roaring-Bitmap bit-sliced indexes, an entity graph with semantic/causal edges, vec2text inversion (four ONNX models) to rebuild forgotten memories, X25519+orthogonal-rotation encryption that preserves cosine similarity, and RTL-SDR radio noise as an entropy/time source for "dream cycles."

LUME is a separate, public, BSD-licensed Rust hybrid search engine (github.com/DeepBlueDynamics/lume) by Steve Harris and Kord Campbell that indexes Markdown, source code, and PDFs using three primitives: field-aware BM25 ($k_1=1.2, b=0.75, \text{title_weight}=2.0$), dense GTR-T5 vectors (768-d, via the "Shivvr" embedding service), and a significance-scored entity co-occurrence graph (a z-score against an independence expectation, log-compressed and tanh-squashed into $[-1,1]$, i.e. Trey Grainger's foreground-vs-background SKG relatedness). The signals fuse multiplicatively: $\text{hybrid} = \text{bm25} \times (1 + \alpha \cdot \text{semantic} + \beta \cdot \text{skg})$. Ferricula and Lume are complementary: **Lume is the document index; Ferricula is the lifecycle/consolidation memory layer**. This briefing maps each to the external frontier.

1. Agent memory architectures

- **Cognitive taxonomy is now standard.** Frameworks explicitly separate short-term/working memory (the context window) from long-term memory, and within long-term memory distinguish **episodic** (specific past interactions), **semantic** (facts/preferences), and **procedural** (learned behaviors/rules) memory — drawn from Tulving's episodic/semantic distinction and the Atkinson–Shiffrin multi-store model.
- **Tiered/OS-style memory (MemGPT → Letta).** MemGPT introduced treating the LLM context window as virtual memory with paging between "core" (RAM-like, in-context), "recall," and "archival" (disk-like) tiers, self-managed by the LLM via tool calls. It is now productized as Letta; independent evals put Letta around ~83% on LoCoMo. (arxiv)
- **Production frameworks (2025–2026):**
 - **Mem0** — hybrid vector+graph+key-value store with automatic fact extraction (ECAI 2025, arXiv:2504.19413, ~47K GitHub stars; raised \$24M in Oct 2025). Per Mem0's April 2026 token-efficient report: "Mem0's new token-efficient memory algorithm hits 92.5 on LoCoMo, 94.4 on LongMemEval, and 64.1/48.6 on BEAM (1M/10M) while averaging under 7,000 tokens per retrieval call. Full-context approaches on the same benchmarks use 25,000+." Largest gains were +29.6 points on temporal queries and +23.1 on multi-hop. (arxiv) (Mem0)
 - **Zep / Graphiti** — temporal knowledge graph tracking fact-validity windows. Per the paper "Zep: A Temporal Knowledge Graph Architecture for Agent Memory" (arXiv:2501.13956): on the DMR benchmark "Zep demonstrates superior performance (94.8% vs 93.4%)," and on LongMemEval (GPT-4o) Zep scores 63.8% vs Mem0's 49.0% — a ~15-point advantage on temporal retrieval. Graphiti OSS has ~27K GitHub stars; Zep is the only major framework with SOC 2 Type 2 + HIPAA + GDPR certification. (arXiv)
 - **A-MEM** (Xu et al., NeurIPS 2025, arXiv:2502.12110) — Zettelkasten-inspired self-organizing memory: atomic notes with LLM-generated keywords/tags/links that evolve as a "living" knowledge graph; a LinkGenerator uses entity co-occurrence, semantic similarity, tag overlap, and LLM relationship reasoning, and memory evolution updates older notes when new ones arrive. (arxiv + 4)

- **Cognee** — Extract-Cognify-Load (ECL) pipeline into a typed knowledge graph (best for local-first, privacy-critical graph reasoning; lacks SOC 2/HIPAA).
- **Generative Agents** (Park et al., 2023) — the memory-stream + reflection + importance/recency/relevance retrieval design that seeded the field. (arxiv)
- **HippoRAG** (Gutiérrez et al.) — neurobiologically inspired, uses Personalized PageRank over a knowledge graph to mimic hippocampal associative recall for multi-hop reasoning.
- **MemoryBank** — adds Ebbinghaus forgetting-curve decay.
- Newer 2025–2026 systems: LightMem (Atkinson–Shiffrin-inspired lightweight pipeline), MemoryOS, EverMemOS (self-organizing memory OS), LiCoMemory (lightweight hierarchical CogniGraph), MemPalace (MIT, reports 88.9% LoCoMo / 96.6% LongMemEval), Hermes Agent (Nous Research, open-sourced Feb 2026), and learned/RL-trained memory controllers (Mem-α, and Mem-T with MoT-GRPO, which improves LoCoMo F1 by ~33%).
- **Consolidation, summarization, forgetting:** recurring patterns include LLM-driven summarization of old turns pushed down the hierarchy, deduplication/consolidation of salient facts (Mem0), forgetting-curve decay (MemoryBank), reversible compression (R³Mem), and reflective memory management. (arxiv)
- **Benchmarks: LoCoMo** (1,540 questions across single/multi-hop/temporal/open-domain, ~600 turns and ~16K tokens over up to 32 sessions); **LongMemEval** (500 questions across five abilities: information extraction, multi-session reasoning, temporal reasoning, knowledge updates, abstention; ~115K-token histories); and **BEAM**, which per Mem0's State of AI Agent Memory 2026 report "evaluates memory systems at 1M and 10M token scales across ten task categories, including preference following, temporal reasoning, and contradiction resolution... the only public benchmark that operates at context volumes production AI agents actually encounter." EpBench targets episodic memory at 100K–1M tokens. Key caveat: vendor-reported scores disagree and most benchmarks test chat history rather than task execution. (Mem0)
- **Self-editing / reflection / memory-as-learning:** Self-RAG reflection tokens, ExpeL (experiential learning), reflective memory management, and RL-trained memory control (MemR³, Mem-T) are the leading directions. Mem0's report names the hardest open problems as "cross-session identity, temporal abstraction at scale, and memory staleness."

2. Thermodynamic and energy-based memory (priority area)

- **Modern Hopfield Networks / Dense Associative Memories (DenseAMs).** Krotov & Hopfield (2016) and Demircigil et al. (2017) introduced energy functions with much higher (exponential/polynomial) storage capacity than the classical Hopfield network. Ramsauer et al. (2021, "Hopfield Networks is All You Need") showed the one-step update of a continuous modern Hopfield network is mathematically equivalent to transformer self-attention — the central bridge between associative memory and attention. Krotov & Hopfield (2021)

generalized this with Lagrangian-defined hierarchical/multi-layer associative memories. 2025–2026 work continues: "In-Context Denoising with One-Layer Transformers" (Smart et al., arXiv:2502.05164) shows a trained attention layer performs one gradient-descent step on a context-aware DenseAM energy landscape (context tokens = associative memories, query = initial state); "On the Role of Hidden States of Modern Hopfield Network in Transformer" (arXiv:2511.20698) develops the hidden-state correspondence further.

- **Diffusion models as associative memory.** Ambrogioni ("In Search of Dispersed Memories," *Entropy* 26(5):381, 2024) proved the low-temperature energy landscape of generative diffusion models trained on discrete patterns is asymptotically identical to that of modern Hopfield networks: data points = memories, noise variance = effective temperature (β^{-1}), reverse process = retrieval. "The Statistical Thermodynamics of Generative Diffusion Models" (Ambrogioni, *Entropy* 27(3):291, 2025) frames generation as symmetry-breaking phase transitions in the energy landscape. "Memorization to Generalization" (Pham et al., NeurIPS 2024, arXiv:2505.21777) shows spurious states and generalization emerge as a phase transition with training-set size. "Dense Associative Memory for Gaussian Distributions" (arXiv:2509.23162) extends DenseAM to probability measures.
- **Stochastic thermodynamics of memory (the most directly relevant frontier paper).** "Stochastic Thermodynamics of Associative Memory" by Spencer Rooke, Dmitry Krotov, Vijay Balasubramanian, and David Wolpert (arXiv:2601.01253, submitted 3 Jan 2026, rev. Apr 2026; cond-mat.stat-mech) presents three results, verbatim: "(1) We establish that dense networks with higher order nonlinearities have a failure mode of pattern completion at nonzero temperatures that is absent from lower order networks and when the temperature is zero; (2) We introduce a method that is exact in the mean field limit for calculating the amount of work expended in these networks under arbitrary fast driving; (3) We use this new method to demonstrate tradeoffs between entropy production, memory retrieval, and operation speed."
 - **The finite-temperature failure mode:** "For $k > 2$ there is always a local minimum of the free energy at zero alignment for any finite temperature, leading to a spurious stored memory." Physically, "the energy landscapes become flatter near zero alignment, but steeper near alignment with a memory, as k increases," so "thermal fluctuations can preferentially walk the state of the system along the flat regions... when the initial state is not well-aligned with a memory." (arxiv) (arxiv)
 - **The central tradeoffs:** "for successful recall we must drive networks more slowly if they are at higher temperature or if the partial memories are more corrupted. At fixed temperature, faster driving additionally incurs higher work cost... The entropy production in this case equals the work cost times the inverse temperature." And: "while higher-order networks have increased storage capacity and better reconstruction accuracy, they incur greater power cost and require stronger control fields. Conversely, lower-order networks are more thermodynamically efficient at low memory loads." Lower-order networks also "have larger basins of attraction... and

hence can perform reconstruction of initial states that are more corrupted... but in exchange will have larger fluctuations/errors in the reconstructions." [arxiv](#)

- **Method:** Total irreversible entropy production is $\Delta S_{\text{tot}} = \beta(W - \Delta F) \geq 0$; the authors state "to our knowledge we are presenting the first application of DMFT [dynamical mean field theory] to calculate work and entropy production in nonstationary, out-of-equilibrium neural network processes," with the instantaneous mean-field power density (their "most important result") depending on the *history* of the macroscopic alignment state, exact at large N for load $p \ll N^{(k-1)}$. **Caveat:** results are restricted to low-to-intermediate load and a single, explicitly non-optimal illustrative control family; optimal control and near-saturation regimes are deferred. A companion, "Geometric Entropy and Retrieval Phase Transitions in Continuous Dense Associative Memory" (arXiv:2604.07401), derives finite-temperature phase boundaries under exponential load.
- **Free-energy principle / active inference (Friston).** Active inference casts agents as minimizing variational free energy against a generative model; memory appears as hierarchical/deep generative state representations and as cognitive maps (e.g., Free Energy Projective Simulation, Pazem et al., *PLOS One* 2025, which explicitly relates FEP to Boltzmann machines and simulated annealing). 2025 work formalizes scalable synthetic active inference via message passing on Forney-style factor graphs (van de Laar, Koudahl & de Vries, *Neural Computation* 37(1), 2025).
- **Equilibrium propagation & Boltzmann machines.** Equilibrium Propagation (Scellier & Bengio, 2017) trains energy-based models via local, steady-state dynamics — hardware-friendly, computing gradients through continuous-time relaxation with only local interactions. 2025–2026 extensions: spiking (StochEP, with mean-field guarantees), quantum (QEP via Onsager reciprocity), oscillator Ising machines, nonlinear-wave media, and analog/memristor in-memory learning; FlowEqProp (arXiv:2604.08150) marries EP with flow matching. Boltzmann machines connect to attention via "Attention in a family of Boltzmann machines emerging from modern Hopfield networks" (arXiv:2212.04692). The 2026 survey "Energy-Based Dynamical Models for Neurocomputation, Learning, and Optimization" (arXiv:2604.05042) ties these together. [arXiv](#)
- **Thermodynamic computing hardware. Extropic unveiled** (Oct 2025) Thermodynamic Sampling Units (TSUs) using probabilistic bits (p-bits) that sample directly from energy-based models. Per Extropic's "Thermodynamic Computing: From Zero to One," "Simulations of DTMs [Denoising Thermodynamic Models] running on TSUs could be 10,000x more energy-efficient than modern algorithms running on GPUs," with each p-bit "generating millions of biased random bits per second using 10,000× less energy than a GPU's floating-point addition." The XTR-0 dev platform has shipped; the Z-1 production chip ($\approx 250,000$ p-bits/chip) is slated for early 2026; the open-source "thrml" library enables simulation. **Normal Computing** (founded 2022; CEO Faris Sbahi, Chief Scientist Patrick Coles, with fluctuation-theorem namesake Gavin Crooks on staff) taped out the **CN101**, announced Aug 12, 2025 as "the world's first thermodynamic computing chip... achieving up to 1000×

energy consumption efficiency on targeted AI and scientific workloads." The Carnot architecture has "four compute tiles containing 64 state variables each, and a reconfigurable NoC... able to support 256-dimensional problems with 32-bit state variables," with a roadmap to CN201/CN301. Its peer-reviewed demonstration ("Thermodynamic computing system for AI applications," *Nature Communications* 16, 2025) used eight RLC circuits for Gaussian sampling and matrix inversion. Normal was selected for ARIA's £50M Scaling Compute Programme (Oct 2024) and raised \$50M led by Samsung Catalyst Fund (March 2026, >\$85M total). Foundational paper: "Thermodynamic AI and the fluctuation frontier" (Coles et al., 2023) treats noise as a computational resource, unlike quantum computing "where noise is a roadblock." Both companies treat **sampling from energy-based distributions** — the same math underlying associative memory — as the native hardware operation. (Pjfp + 3)

- **Landauer / information thermodynamics.** Landauer's bound ($k_B \cdot T \cdot \ln 2$ per bit erased, $\approx 3 \times 10^{-21}$ J at 300 K) frames the energy cost of forgetting; 2026 experimental work measured deviations from it in silicon DRAM cells (*Phys. Rev. Lett.* 136, 117103, 2026), finding the Landauer limit is not reached even under infinite-time erasure. This is the physical grounding for treating decay/forgetting as a thermodynamically meaningful operation, as Ferricula does. (arxiv) (APS Journals)

3. Agentic document indexing (for LUME)

- **Hybrid retrieval is the frontier default:** lexical (BM25) + dense (embeddings) + graph signals, fused. Lume's multiplicative blend with significance-scored (not raw co-occurrence) entity edges — which resists "promiscuous hubs" and a regression test confirms outranks a perfect-Jaccard hub — is a sophisticated, auditable instance of this.
- **Graph & hierarchical RAG:** Microsoft GraphRAG (entity-relation graphs), RAPTOR (recursive summarization into hierarchical trees for multi-level retrieval), LightRAG (KG + vector, local+global search), HippoRAG (Personalized PageRank associative recall). Agentic RAG (Self-RAG, PlanRAG, Search-R1, Search-o1, A-RAG) adds autonomous plan/retrieve/reflect loops; A-RAG reports 94.5% on HotpotQA and 89.7% on 2WikiMultiHop, and Self-RAG achieved a 5.8% hallucination rate (lowest among 12 variants in a 2025 MDPI study). The GA-RAG survey (arXiv:2501.09136 and successors) formalizes graph-based agentic RAG patterns (model-mediated traversal, evaluator-optimizer loops, hybrid retrieval).
- **Chunking:** the spectrum runs fixed-size → recursive → semantic (embedding-similarity breakpoints) → LLM-based (LumberChunker, Meta-Chunking via perplexity) → **agentic** (the agent picks strategy per document) → late chunking (embed full doc with a long-context model, then split, so each chunk carries long-range context) → hierarchical. Page-level chunking won NVIDIA's 2024 benchmarks for paginated documents (0.648 accuracy, lowest variance); semantic chunking can improve recall by up to ~9% but is costly and sometimes unstable on real documents. New metrics (Boundary Clarity, Chunk Stickiness) and chunk-free pipelines (CFIC, M-RAG) are emerging.

- **Embedding models (2025-2026):** Per the Qwen3 Embedding release (arXiv:2506.05176), "The 8B size embedding model ranks No.1 in the MTEB multilingual leaderboard (as of June 5, 2025, score 70.58)"; it supports 100+ languages, 32K context, and flexible (Matryoshka) dimensions from 32 to 4096 under Apache-2.0, with 0.6B/4B/8B variants. BGE-M3 (multi-functional dense+sparse+multi-vector, 100+ languages, MIT) is the production workhorse; NV-Embed-v2, Gemini-Embedding (multimodal, 3072-d), Cohere embed-v4, voyage-4, and Jina v5 are other leaders. Lume's GTR-T5 (768-d) is now dated; Qwen3-Embedding or BGE-M3 would be upgrades. (Qwen + 2)
- **Late interaction / multi-vector:** ColBERT (token-level MaxSim late interaction, Khattab & Zaharia 2020) → ColBERTv2/PLAID/WARP → **ColPali/ColQwen** (Faysse et al., ICLR 2025; visual document retrieval over image patches via vision-language models; SigLIP + PaliGemma/Qwen producing 128-d patch embeddings; evaluated on the ViDoRe benchmark by NDCG@5). Efficiency variants: ColModernVBERT (250M params, within 0.6 pts of ColPali), token-pruning methods. This is the biggest capability Lume currently lacks and the natural path for indexing visual/PDF documents.
- **Embedding inversion (vec2text):** Morris et al.'s vec2text reconstructs text from embeddings via iterative hypothesis refinement (reported ~97 BLEU and ~92% recovery for 32-token inputs). Ferricula uses exactly this technique (four ONNX models: embedder, projection, T5 encoder, decoder) to "reconstruct forgotten memories from their vector seeds" — a creative, frontier-aware repurposing of a method normally framed as a privacy attack. Note the literature shows quantization and noise injection degrade inversion fidelity, relevant to Ferricula's encryption/decay interactions.

4. Synthesis — where the threads intersect

The frontier intersection of agent memory + thermodynamics + document indexing is real and underexplored, which is precisely the niche Ferricula/Ferricula occupies:

- **Energy landscapes unify memory and generation.** DenseAMs, attention, and diffusion are all the same energy-based retrieval dynamics at different temperatures (Ramsauer; Ambrogioni). A memory system whose storage/retrieval is framed as relaxation on an energy landscape (Ferricula's "fidelity as energy state") is conceptually continuous with this literature.
- **Decay/forgetting as thermodynamics.** The Rooke-Krotov-Balasubramanian-Wolpert tradeoff (slower driving needed at higher temperature / more corruption; entropy production = $\beta \times \text{work}$) gives a principled vocabulary for Ferricula's decay- α , dream-cycle consolidation, and "fidelity gate." The Landauer framing makes "forgetting has an energy cost" literally true. Notably, Krotov is a co-author of the frontier paper *and* the originator of modern DenseAMs — the same intellectual lineage Ferricula invokes.
- **Hybrid exact retrieval over a pruned set.** Ferricula's choice of exact cosine over a bitmap-pre-filtered candidate set (rejecting ANN because "an agent's memory is constantly decaying, consolidating, and pruning — exactly the workload ANN indexes handle worst,"

and because at 95% recall@10 "that 5% miss rate is a lie the agent tells itself") is a defensible, frontier-aware engineering position given small active-set sizes ($\leq 50K$ memories ≈ 146 MB; $\sim 200\mu s$ with bitmap pre-filter).

Details: Open problems and promising directions

1. **Cross-session identity, temporal abstraction at scale, and memory staleness** are named by practitioners (Mem0 2026 report) as the hardest open problems; BEAM is deliberately designed so no current system saturates it (best scores $\sim 64\%/49\%$ at 1M/10M tokens).
2. **Memory capacity scaling laws via statistical physics.** DenseAM capacity scales as $\alpha \cdot k \cdot N^{(k-1)}$; the phase-transition/spin-glass view (Amit-Gutfreund-Sompolinsky 1985; Lucibello-Mézard 2024; "Geometric Entropy and Retrieval Phase Transitions," arXiv:2604.07401) gives capacity and retrieval-stability bounds under entropic penalty — a lens for sizing an agent's "active memory" budget against a target recall accuracy.
3. **Optimal control of decay/consolidation.** The 2026 stochastic-thermodynamics paper explicitly defers optimal driving protocols; an agent memory system could in principle *learn* decay/dream schedules that minimize entropy production for a target recall accuracy — a concrete, novel research program Ferricula is positioned to attempt.
4. **Late-interaction + graph fusion for documents.** Combining ColPali-style visual late interaction with significance-weighted entity graphs (Lume's strength) is not yet standard practice.
5. **Self-organizing + thermodynamic.** A-MEM's Zettelkasten self-organization combined with an explicit energy/decay lifecycle (Ferricula) is a novel, unpublished combination worth formalizing and benchmarking.

Open-source tools/repos worth tracking

Letta (MemGPT), Mem0, Graphiti (Zep), A-MEM (WujiangXu/A-mem), Cognee, HippoRAG, RAPTOR, Microsoft GraphRAG, LightRAG; vec2text (jxmorris12/vec2text); Extropic thrml; Normal Computing thermox; ColBERT/PLAID/WARP, ColPali/ColQwen. Benchmarks to adopt: LoCoMo, LongMemEval, BEAM, EpBench (memory); ViDoRe (visual document retrieval); BEIR/MTEB (embedding/retrieval).

Recommendations

1. **Anchor LUME's retrieval on the proven hybrid stack it already has, but modernize embeddings.** Swap GTR-T5 (768-d) for Qwen3-Embedding (configurable 1024-4096-d, Apache-2.0, MTEB-multilingual #1 at 70.58) or BGE-M3 (MIT, dense+sparse+multi-vector); keep the BM25 + significance-graph blend, which is genuinely frontier-grade and auditable. **Threshold to change:** if recall@k on your own corpus is materially below a Qwen3/BGE-M3 baseline, switch; otherwise the engineering cost of re-indexing isn't justified.

2. **Add late-interaction / multi-vector retrieval for documents and PDFs.** Adopt a ColBERT-style index (PLAID/WARP) for text and ColPali/ColQwen for visual/PDF pages — this is LUME's largest capability gap versus the frontier. **Benchmark on ViDoRe**; proceed if NDCG@5 beats the current dense+BM25 blend on your document mix.
3. **Adopt agentic + hierarchical chunking and recursive summarization (RAPTOR-style) for large docs**, with page-level chunking for paginated PDFs. Apply the benchmark-pair discipline (always pair an accuracy number with its token cost) the Mem0 2026 report recommends.
4. **Formalize Ferricula's thermodynamic claims against the literature.** Cite Rooke et al. (arXiv:2601.01253), Ambrogioni (diffusion-as-AM), and Ramsauer et al. (Hopfield = attention) to ground "memory as entropy" — and be precise in any whitepaper that current thermodynamic *hardware* (Extropic, Normal Computing) is early-stage and the 10,000×/1000× figures are vendor simulation/target claims, not benchmarked results.
5. **Evaluate Ferricula/Lume on standard memory benchmarks.** Run LoCoMo, LongMemEval, and (aspirationally) BEAM so the project's numbers are comparable to Mem0 (92.5/94.4), Zep (63.8 LongMemEval), and Letta (~83 LoCoMo). Without this, the system's quality is unverifiable relative to the field — this is the single highest-leverage credibility step.
6. **Treat decay/consolidation as a controllable thermodynamic schedule.** The entropy-production/accuracy/speed tradeoff implies dream-cycle cadence and decay- α should be tuned (or learned) per target recall level rather than fixed — the most defensible way to turn the thermodynamic metaphor into a measurable advantage.

Caveats

- **"Furricula" vs "Ferricula":** the public project is spelled Ferricula; I proceeded on the assumption these are the same project. The **Ferricula source repo does not appear public** — only the ferricula.com marketing/whitepaper site and the public Lume repo (plus the org's other repos: hyperia, nemesis8, shivvr, grubcrawler, charon, sdr-random) were accessible. Detailed internal claims (decay curves, dream cycles, SDR entropy, encryption-preserving cosine similarity) come from the project's own site and whitepaper and are **not independently verified**.
- **Vendor-reported benchmark scores conflict.** Mem0 and Zep report materially different LongMemEval numbers for each other (e.g., Mem0 at 49.0% per Zep vs Mem0's own higher self-reported figures); treat all single-source scores as indicative, not settled, and re-run evals on your own workload.
- **Thermodynamic-hardware claims are early and promotional.** Extropic's ~10,000× and Normal Computing's ~1000× figures are company claims on targeted/simulated workloads with early test chips, not peer-reviewed head-to-head benchmarks. The peer-reviewed Normal Computing demonstration used only eight RLC circuits.

- **The strongest theory-to-system links are conceptual analogies, not proven equivalences.** Diffusion = Hopfield holds asymptotically under specific (discrete-pattern, low-temperature) conditions; the stochastic-thermodynamics tradeoffs are proven for specific control families and load regimes ($p \ll N^{(k-1)}$). Ferricula's "thermodynamic memory" is a well-motivated metaphor backed by real frontier mathematics, **not** a literal physical thermodynamic computer.
- **Date-stamps:** several arXiv IDs cited here carry 2026 prefixes (2601.xxxxx, 2602.xxxxx, 2603.xxxxx, 2604.xxxxx) consistent with the mid-2026 research horizon; readers collating with the parallel Gemini run should verify exact version/publication status, as some are recent preprints that may still be under revision.